

Inflationary potentials to implement in COSMOLATTICE

in light of *Planck* 2018, BICEP/Keck 2018, ACT DR6 and SPT-3G

Working notes based on J. Ellis, M. A. G. Garcia, K. A. Olive, S. Verner,
Constraints on Attractor Models of Inflation and Reheating
from *Planck*, *BICEP/Keck*, *ACT DR6*, and *SPT-3G* Data, [arXiv:2510.18656](https://arxiv.org/abs/2510.18656)

October 2, 2026

Abstract

We list the single-field inflaton potentials that remain viable after the latest CMB data and that should be implemented as COSMOLATTICE models for lattice studies of (p)reheating. For each family we give the potential, its first and second field derivatives (needed for `potentialTerms`, `potDeriv`, `potDeriv2`), the CMB normalization, the end-of-inflation field value, the parameter ranges allowed by the data, and a suggested choice of program variables (f_* , ω_* , α_t). All derivative expressions have been checked numerically.

Contents

1	Observational targets	1
2	Conventions	2
3	Models to implement	3
3.1	Model 1: generalized E-model (α -Starobinsky)	3
3.2	Model 2: generalized T-model	3
3.3	Model 3: deformed E-model	4
3.4	Model 4: deformed T-model	4
4	Benchmark points from the data	4
5	Implementation notes for CosmoLattice	5
5.1	Program variables	5
5.2	Initial conditions	6
5.3	Daughter fields and reheating	6
5.4	Numerical caveats	6
5.5	Overlap with existing models	6
6	Other viable families (outside the scope of the paper)	7

1 Observational targets

The three n_s determinations used to discriminate between models are (68% CL, pivot $k_* = 0.05 \text{ Mpc}^{-1}$):

$$\textit{Planck} \text{ 2018 (+lensing)} : n_s = 0.9649 \pm 0.0042, \quad (1)$$

$$\text{CMB-SPA (SPT-3G + \textit{Planck} + ACT)} : n_s = 0.9684 \pm 0.0030, \quad (2)$$

$$\text{P-ACT-LB2 (\textit{Planck} + ACT + lensing + DESI DR2)} : n_s = 0.9752 \pm 0.0030, \quad (3)$$

together with the tensor bound (95% CL)

$$r_{0.05} < 0.036 \quad (Planck+BK18), \quad r < 0.038 \quad (Planck+ACT+BK18), \quad (4)$$

and the scalar amplitude $A_s \simeq 2.10 \times 10^{-9}$. The P-ACT-LB2 value sits $\sim 2\sigma$ above *Planck*; CMB-SPA is in between. A model is therefore “safe” if it reaches $n_s \gtrsim 0.968$ with $r \lesssim 0.036$ for a physically reasonable reheating history, and “ACT-preferred” if it reaches $n_s \simeq 0.972$ – 0.978 .

Reheating sets the number of e -folds N_* ; the paper scans $4 \text{ MeV} \leq T_{\text{RH}} \leq 2 \times 10^{15} \text{ GeV}$, with the “conservative” window $100 \text{ GeV} \lesssim T_{\text{RH}} \lesssim 10^{10} \text{ GeV}$ (leptogenesis/sphalerons below, gravitino bound above).

Consequences for model selection.

- Chaotic monomials $V \propto \varphi^2, \varphi^4$ are excluded by r (already at $> 3\sigma$ for φ^4 with *Planck*+BK15). They should only be kept as toy preheating models (`lphi4`, `m2phi2`), not as realistic inflationary models.
- The canonical Starobinsky/T-model ($\alpha = 1$, $k = 2$) are fine for *Planck* but sit below the CMB-SPA 95% region and well below P-ACT-LB2.
- Plateau models with a *steeper minimum* ($V \propto \varphi^k$, $k \geq 6$) or a *slightly deformed plateau* ($\kappa \simeq 0.9999$) raise n_s into the ACT/SPA range. These are the main new targets, and their reheating stage (anharmonic oscillations, fragmentation) is exactly what a lattice code is needed for.

2 Conventions

We use $x \equiv \varphi/M_P$, $M_P = 2.435 \times 10^{18} \text{ GeV}$, and

$$V_0 \equiv \frac{3}{4} \lambda M_P^4, \quad \beta \equiv \sqrt{\frac{2}{3\alpha}}, \quad \gamma \equiv \frac{1}{\sqrt{6\alpha}} = \frac{\beta}{2}. \quad (5)$$

Here α is the attractor parameter (curvature of the no-scale Kähler manifold, $R_K = 2/3\alpha$); $k \geq 2$ is an even integer setting $V \propto \varphi^k$ near the minimum; κ is the deformation parameter ($\kappa = 1$: undeformed). Primes denote d/dx , so $\partial_\varphi V = V'/M_P$ and $\partial_\varphi^2 V = V''/M_P^2$.

Name clash. COSMOLATTICE already uses the member `alpha` for the time-rescaling exponent ($d\tilde{\eta} = a^{-\alpha_t} \omega_* dt$). In the code the attractor parameter must be named differently, e.g. `alphaAtt`. In these notes the time-rescaling exponent is denoted α_t .

CMB normalization. To good approximation, for all families below with $\kappa = 1$,

$$\lambda \simeq \frac{24 \alpha \pi^2 A_s}{N_*^2} \quad \Rightarrow \quad \lambda \simeq 1.64 \times 10^{-10} \alpha \left(\frac{55}{N_*} \right)^2, \quad (6)$$

independent of k . For deformed models ($\kappa \neq 1$) and for precision work, λ should be fixed numerically from $A_s = V_*/(24\pi^2 \varepsilon_{V*} M_P^4)$ or, better, by solving the Mukhanov–Sasaki equation.

Slow-roll and end of inflation. $\varepsilon_V = \frac{1}{2}(V'/V)^2$, $\eta_V = V''/V$, $n_s \simeq 1 - 6\varepsilon_{V*} + 2\eta_{V*}$, $r \simeq 16\varepsilon_{V*}$. Inflation ends at $\varepsilon_H = 1$, i.e. $\varepsilon_V \simeq (1 + \sqrt{1 - \eta_V/2})^2$. In the large- N_* limit all undeformed attractors give

$$n_s \simeq 1 - \frac{2}{N_*}, \quad r \simeq \frac{12\alpha}{N_*^2}. \quad (7)$$

3 Models to implement

Table 1 summarizes the proposed model files. Models are grouped so that each family can be one COSMOLATTICE header with $(\alpha, k, \kappa, \lambda)$ read from the parameter file; the separate entries are the benchmark points worth running.

Priority	Proposed file	Potential	Parameters	Status w.r.t. data
1	<code>attractorE.h</code> ($\kappa = 1$)	gen. E-model, Eq. (8)	α, k, λ	$k \geq 6$: improved n_s ; $\alpha \lesssim 16$
1	<code>attractorT.h</code> ($\kappa = 1$)	gen. T-model, Eq. (14)	α, k, λ	$k = 10$, $\alpha \simeq 1$ enters ACT 95%
1	<code>attractorE.h</code>	deformed E, Eq. (20)	$\alpha, k, \kappa, \lambda$	$\kappa \simeq 1 - 10^{-4}$: fits P-ACT-LB
1	<code>attractorT.h</code>	deformed T, Eq. (24)	$\alpha, k, \kappa, \lambda$	$\kappa \simeq 1 - 10^{-4}$: fits all three sets
2	<code>attractorE.h</code> ($k = 2, \kappa = 1$)	E-model / Starobinsky	α, λ	<i>Planck</i> OK; $\alpha \lesssim 25$
2	<code>attractorT.h</code> ($k = 2, \kappa = 1$)	T-model	α, λ	<i>Planck</i> OK; $\alpha \lesssim 11$

Table 1: Implementation list. Since the deformed models reduce to the generalized ones at $\kappa = 1$, two headers cover all families: `models/attractorE.h` and `models/attractorT.h`, with $(\alpha, k, \kappa, \lambda, g)$ read at run time. The $k = 2$ undeformed cases are obtained by setting $k = 2, \kappa = 1$; the existing `tanh2.h` already covers the $k = 2$ T-model in a different parametrization (Sec. 5.5).

3.1 Model 1: generalized E-model (α -Starobinsky)

$$V(\varphi) = V_0 \left(1 - e^{-\beta x}\right)^k. \quad (8)$$

For $\alpha = 1, k = 2$ this is the Starobinsky model (Einstein frame of $R + R^2/6M_P^2$, with $\lambda = M^2/M_P^2$). With $u \equiv 1 - e^{-\beta x}$:

$$V' = V_0 k \beta e^{-\beta x} u^{k-1}, \quad (9)$$

$$V'' = V_0 k \beta^2 e^{-\beta x} u^{k-2} \left(k e^{-\beta x} - 1\right). \quad (10)$$

Small field: $V \simeq V_0 \beta^k x^k$; for $k = 2$, $m_\varphi^2 = \lambda M_P^2/\alpha$. Note the potential is asymmetric: for $x < 0$ it grows exponentially, so oscillations are not symmetric about the minimum.

End of inflation (slow-roll estimate, accurate to a few %):

$$x_{\text{end}} \simeq \sqrt{\frac{3\alpha}{2}} \ln \left[\frac{2 \left(6\alpha + \sqrt{3\alpha(2k-1)^2 - k}\right)}{12\alpha - 1} \right], \quad (11)$$

giving $x_{\text{end}} \simeq 0.63, 1.16, 1.52$ for $\alpha = 1$ and $k = 2, 4, 6$.

Horizon exit:

$$x_* \simeq \sqrt{\frac{3\alpha}{2}} \left[1 + \frac{3\alpha}{2kN_* - 3\alpha} \right] \ln \left(\frac{2kN_*}{3\alpha} + e^{\sqrt{2/3}x_{\text{end}}} - \sqrt{\frac{2}{3}}x_{\text{end}} \right). \quad (12)$$

Observables (large N_*):

$$n_s \simeq 1 - \frac{2}{N_*} - \frac{3\alpha(2 + k - 2 \ln \frac{2kN_*}{3\alpha})}{2kN_*^2}, \quad r \simeq \frac{12\alpha}{N_*^2} - \frac{36\alpha^2 \ln \frac{2kN_*}{3\alpha}}{kN_*^3}. \quad (13)$$

3.2 Model 2: generalized T-model

$$V(\varphi) = V_0 \tanh^k(\gamma x). \quad (14)$$

With $t \equiv \tanh(\gamma x)$, $s^2 \equiv \text{sech}^2(\gamma x)$:

$$V' = V_0 k \gamma t^{k-1} s^2, \quad (15)$$

$$V'' = V_0 k \gamma^2 t^{k-2} s^2 [(k-1)s^2 - 2t^2]. \quad (16)$$

Small field: $V \simeq V_0 \gamma^k x^k$; for $k = 2$, $m_\varphi^2 = \lambda M_P^2/(4\alpha)$. The potential is symmetric.

End of inflation:

$$x_{\text{end}} \simeq \sqrt{\frac{3\alpha}{2}} \ln \left[\frac{2k(1 - 2\sqrt{3\alpha})}{1 - 12\alpha} + \sqrt{\frac{(1 + 12\alpha - 4\sqrt{3\alpha})(12\alpha + 4k^2 - 1)}{(12\alpha - 1)^2}} \right], \quad (17)$$

($x_{\text{end}} \simeq 0.89$ for $\alpha = 1$, $k = 2$). *Horizon exit:*

$$x_* \simeq \sqrt{\frac{3\alpha}{2}} \cosh^{-1} \left[\frac{2kN_*}{3\alpha} + \cosh(\beta x_{\text{end}}) \right]. \quad (18)$$

Observables:

$$n_s \simeq 1 - \frac{2}{N_*} - \frac{3(k-2)\alpha}{2kN_*^2}, \quad r \simeq \frac{12\alpha}{N_*^2} - \frac{36\alpha^2}{kN_*^3}. \quad (19)$$

3.3 Model 3: deformed E-model

$$V(\varphi) = V_0 [\kappa - \kappa \cosh(\beta x) + \sinh(\beta x)]^k. \quad (20)$$

For $\kappa = 1$ the bracket equals $1 - e^{-\beta x}$ and Eq. (8) is recovered. For $\kappa < 1$ the plateau is lifted (V grows like $[(1 - \kappa)/2]^k e^{k\beta x}$ at large x), which increases n_s and r . With

$$G \equiv \kappa(1 - \cosh \beta x) + \sinh \beta x, \quad G' = \beta(\cosh \beta x - \kappa \sinh \beta x), \quad G'' = \beta^2(\sinh \beta x - \kappa \cosh \beta x), \quad (21)$$

$$V' = V_0 k G^{k-1} G', \quad (22)$$

$$V'' = V_0 k [(k-1)G^{k-2} G'^2 + G^{k-1} G'']. \quad (23)$$

Near the minimum $G \simeq \beta x$, so the small-field limit, m_φ and the reheating stage are the same as for Model 1. The deformation matters only on the plateau ($x \gtrsim 5-6$ for $1 - \kappa \sim 10^{-4}$), i.e. for the CMB observables, not for the lattice dynamics, except through λ and N_* .

3.4 Model 4: deformed T-model

$$V(\varphi) = \frac{V_0}{4} [1 + \kappa - (\kappa - 1) \cosh(\beta x)]^2 \tanh^k(\gamma x). \quad (24)$$

For $\kappa = 1$ the prefactor is V_0 and Eq. (14) is recovered. Writing $H \equiv 1 + \kappa + (1 - \kappa) \cosh \beta x$, $H' = (1 - \kappa)\beta \sinh \beta x$, $H'' = (1 - \kappa)\beta^2 \cosh \beta x$, and $T_k \equiv t^k$ with T'_k, T''_k given by the T-model expressions above divided by V_0 :

$$V' = \frac{V_0}{4} [2HH' T_k + H^2 T'_k], \quad (25)$$

$$V'' = \frac{V_0}{4} [2(H'^2 + HH'') T_k + 4HH' T'_k + H^2 T''_k], \quad (26)$$

with $T'_k = k\gamma t^{k-1} s^2$ and $T''_k = k\gamma^2 t^{k-2} s^2 [(k-1)s^2 - 2t^2]$. At the minimum $H \rightarrow 2$, so the small-field limit matches Model 2.

4 Benchmark points from the data

Table 2 collects the parameter points and limits quoted in arXiv:2510.18656 (exact numerical n_s, r). These are the recommended runs.

Model	k	κ	α	N_* / T_{RH}	(n_s, r)	Comment
E (Starobinsky)	2	1	1	$T_{\text{RH}} = 10^2\text{--}10^{10}$ GeV	$0.958\text{--}0.963, \simeq 0.0035$	<i>Planck</i> OK, below SPA/ACT
E	2	1	≤ 25			upper limit from r
T	2	1	1	same	$0.956\text{--}0.961$	tension with <i>Planck</i>
T	2	1	≤ 11			upper limit from r
E	4	1	1	$N_* \simeq 55.7$ (any T_{RH})	$\simeq 0.965$	$w = 1/3$, N_* indep. of T_{RH} ; $\alpha \lesssim 17$
E	6	1	1	$N_* \simeq 56\text{--}58$	$\simeq 0.965$	$\alpha < 16$
E	6	1	$\simeq 5$	$N_* \simeq 58$	$\simeq 0.966$	touches ACT 95%
E	10	1	$\simeq 5$			overlaps ACT and SPA
T	4	1	$\lesssim 11$			<i>Planck</i> OK, below ACT
T	6	1	1	$N_* \simeq 58$	$\simeq 0.965$	
T	10	1	1	$N_* \simeq 59\text{--}61.5$	$0.965\text{--}0.967$	enters ACT 95%
deformed E	2	0.9999	1	$T_{\text{RH}} = 10^2\text{--}10^{10}$ GeV	$0.968\text{--}0.975, 0.007\text{--}0.005$	fits P-ACT-LB; $\alpha \leq 9$
deformed T	2	0.9999	1	same	$0.970\text{--}0.977$	fits all three; $\alpha \leq 11$
deformed E	4	0.99999 [†]	1		$\simeq 0.970, \simeq 0.004$	fits all three; $\alpha \leq 15$
deformed E	6	0.9999	4	$N_* = 56\text{--}58$	$\simeq 0.974, \simeq 0.016$	fits all three; $\alpha \leq 16$
deformed T	4	0.99999 [†]	1		$\simeq 0.966, \simeq 0.004$	SPA OK, marginal ACT
deformed T	6	0.99999 [†]	1		$\simeq 0.969, \simeq 0.012$	overlap of all three

Table 2: Benchmark points and upper limits on α from arXiv:2510.18656. Empty cells: not quoted in the paper. [†]The text of the paper says $\kappa = 0.9999$ for $k = 4, 6$, but the figure file is named `fulltplots_kappa0p99999.pdf`, and our background solver (`notes/attractor_ics.py`) reproduces the quoted n_s, r only for $\kappa = 0.99999$; with $\kappa = 0.9999$ these points give a blue tilt ($n_s = 0.99\text{--}1.04$). The deformed E-model $k = 6$, $\alpha = 4$ point does match $\kappa = 0.9999$ ($n_s = 0.973$, $r = 0.017$). The deformed T $k = 6$ value $r \simeq 0.012$ is not reproduced for either κ (we get $r \simeq 0.004$).

Recommended minimal set of runs.

1. E-model $k = 2$, $\alpha = 1$ (Starobinsky): reference; compare with the existing literature.
2. E- and T-models with $k = 6$ and $k = 10$, $\alpha = 1$ and $\alpha = 5$: the reheating dynamics (stiff w , fragmentation) is what drives N_* up and must be computed on the lattice.
3. Deformed E- and T-models, $\kappa = 0.9999$, $k = 2, 4, 6$, $\alpha = 1$ (and $\alpha = 4$ for deformed E, $k = 6$).

5 Implementation notes for CosmoLattice

5.1 Program variables

Following the conventions of the existing models (`lphi4.h`: $\alpha_t = 1$, $\omega_* = \sqrt{\lambda} \varphi_i$; `tanh2.h`: $\alpha_t = 0$, $\omega_* = m$), write the small-field limit as $V \simeq (\lambda_k/k) M_P^{4-k} \varphi^k$, i.e.

$$\lambda_k^{(E)} = k \frac{3}{4} \lambda \beta^k, \quad \lambda_k^{(T)} = k \frac{3}{4} \lambda \gamma^k, \quad (27)$$

and choose

$$f_* = M_P, \quad \omega_* = \sqrt{\lambda_k} M_P x_i^{(k-2)/2}, \quad \alpha_t = \frac{3(k-2)}{k+2}, \quad (28)$$

where x_i is the initial field value. The choice of α_t makes the oscillation frequency of a homogeneous condensate constant in program time ($\alpha_t = 0$ for $k = 2$, $\alpha_t = 1$ for $k = 4$, $\alpha_t = 3/2$ for $k = 6$, $\alpha_t = 2$ for $k = 10$). With $\tilde{\varphi} = x$, the program potential $\tilde{V} = V/(f_*^2 \omega_*^2)$ is

$$\tilde{V}_E(x) = \frac{F_E(x)}{k \beta^k x_i^{k-2}}, \quad \tilde{V}_T(x) = \frac{F_T(x)}{k \gamma^k x_i^{k-2}}, \quad (29)$$

with $F \equiv V/V_0$ the shape functions of Sec. 3 (same normalization for the deformed versions). For $k = 2$: $\tilde{V}_E = \frac{3\alpha}{4}F_E$, $\tilde{V}_T = 3\alpha F_T$. Program derivatives are simply $\tilde{V}'(x)$, $\tilde{V}''(x)$ with the same prefactor. λ then enters only through ω_* (the time unit), as for `lphi4`.

Caveat for $k \geq 6$: once the condensate fragments the effective equation of state drops to $w \rightarrow 1/3$, and $\alpha_t = 1$ becomes the natural choice at late times. Check energy conservation and the time step for both choices.

5.2 Initial conditions

- Start at (or slightly before) the end of inflation, $x_i \simeq x_{\text{end}}$ from the formulas above, with the initial velocity from a background (homogeneous) integration of $\ddot{\varphi} + 3H\dot{\varphi} + \partial_\varphi V = 0$ from x_* down to x_i . Slow-roll velocities underestimate $|\dot{\varphi}|$ at $\varepsilon_H \sim 1$.
- `notes/attractor_ics.py` does this: given (model, α , k , κ , N_*) it integrates the background in e -folds, finds x_{end} ($\varepsilon_H = 1$) and x_* , fixes λ from A_s , and prints `lambda`, `initial_amplitudes` and `initial_momenta` at the end of inflation. For Starobinsky with $N_* = 55$ it gives $\lambda = 1.57 \times 10^{-10}$, $x_* = 5.32$, $x_{\text{end}} = 0.615$. N_* is an input; it must be chosen from T_{RH} (Eq. (A.12) of the paper for $k = 2$) or, for $k \geq 6$, from the fragmentation temperature measured on the lattice.
- The benchmark parameter files `models/parameter-files/attractor{E,T}*.in` were generated with it.

5.3 Daughter fields and reheating

The paper treats reheating perturbatively (Γ_φ free, T_{RH} as a parameter). For lattice runs, add as an option:

- quadratic coupling $\frac{1}{2}g^2\varphi^2\chi^2$ (parametric resonance; already used in `tanh2.h`, `lphi4.h`);
- for $k \geq 4$ the inflaton self-resonance alone is the key physics: run first with no daughter field to measure the fragmentation time and T_{frag} , which enter N_* through $\max[T_{\text{RH}}, T_{\text{frag}}]$.

5.4 Numerical caveats

- **Deformed models:** the deformation $1 - \kappa \sim 10^{-4}$ is invisible near the minimum. Use `double`: single precision cannot resolve $1 - \kappa \sim 10^{-4}$ on the plateau. Near the minimum write $G = \sinh \beta x - 2\kappa \sinh^2(\beta x/2)$ to avoid the cancellation in $\kappa(1 - \cosh \beta x)$.
- **Powers of negative bases:** k must be an even integer. Since $u = 1 - e^{-\beta x}$ and $\tanh(\gamma x)$ are negative for $x < 0$, use an integer power (e.g. `pow<k>` with compile-time k , or a template parameter); a floating-point `pow(u, k)` returns NaN for $u < 0$ only when k is non-integer, so never read k as a float.
- **E-model asymmetry:** for $x < 0$, V grows like $e^{2|\beta x|}$ (for $k = 2$); the time step must resolve the steeper side.
- **$\tanh^k$ for large k :** near the minimum $V'' \propto x^{k-2}$ vanishes; the effective mass is field dependent, so initial fluctuation spectra should use the mass at x_i , not at $x = 0$.

5.5 Overlap with existing models

`models/tanh2.h` implements $V = \frac{1}{2}\Lambda^4 \tanh^2(\varphi/M)$, i.e. the $k = 2$ T-model with $M = \sqrt{6\alpha} M_P$ and $\Lambda^4 = \frac{3}{2}\lambda M_P^4$. `models/NMC.tanh4_w_mass_P.h` is a non-minimally coupled $\tanh^4$ -type model. The new headers can be validated against `tanh2.h` by setting $k = 2$, $\kappa = 1$.

6 Other viable families (outside the scope of the paper)

The paper’s introduction lists other models argued to remain compatible with ACT DR6; they are candidates for later, but their data fits were not checked here:

- chaotic/polynomial inflation with non-minimal coupling $\xi\varphi^2 R$ (giving $r \sim 10^{-2}$); `NMC_lphi4.h` is a starting point;
- Starobinsky with higher-curvature corrections (R^3 , etc.) and radiatively corrected attractors;
- polynomial (hilltop/inflection-point) potentials and smooth hybrid inflation ($n_s \simeq 0.97$).